WORKPULSE

System Specification Summary

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1. Project Overview and Users

WorkPulse is an implemented web-based employee attendance and work analytics information system for MVL. It centralizes employee, attendance, leave, cash advance, and payroll records. Interviews identified manual Excel payroll preparation and possible proxy attendance as principal concerns, alongside monitoring, transparency, and attendance inaccuracies.

User or classification	Capabilities
Administrator / Owner	Authorized administrator accounts maintain records and settings, review attendance and corrections, decide requests, record disbursements and payroll settlement, inspect reports, logs, and analytical indicators
Employee	View authorized personal attendance, payroll/payslip summaries, requests, balances, and notifications; submit eligible leave/cash advance requests and maintain permitted contact details
Regular, Extra, Manager / Supervisor	Employee classifications, not separate portals or administrator privileges. Manager and legacy supervisor records use the combined management classification/rate

Implemented Modules

Module	Important functions
Accounts and employee directory	Authentication, account status, archive/access controls, employee records, password changes and recovery
Attendance and biometrics	Configured fingerprint kiosk, manual attendance, daily/date-range views, correction reasons/history, notifications, and settled-payroll protection
Leave and cash advances	Requests and administrator decisions; cash-given confirmation, outstanding balances, and payroll repayment tracking
Payroll and reporting	Attendance-based earnings, additions, carried unpaid earnings, deductions, settlement/history, payslip summaries, CSV/Excel exports
Insights and settings	Statistical attendance forecast, historical absence indicators, unusual-arrival analysis, biometric operational indicators, schedules/rates, audit logs
Offline and recovery	Selected cached views and queued actions; downloadable database backup and separate recovery tools

2. Architecture and Technology

Component	Implementation and boundary
Browser frontend	React 19, TypeScript, Vite, Tailwind CSS, Recharts, and ExcelJS. Administrator, employee, and attendance kiosk interfaces
Application backend	Node.js and Express with MongoDB/Mongoose libraries. Enforces sessions, roles, record ownership, input/business rules, notifications, and audit logging
Database	MongoDB Atlas used through the backend; browser code calls APIs rather than directly connecting to Atlas
Hosting	Vercel frontend and Render backend on free tiers. vercel.json forwards /api requests to the configured Render URL
Local biometric runtime	Compatible DigitalPersona reader/SDK components and the Windows PowerShell/FingerJet helper. The matching code launches powershell.exe where that backend process runs; remote Render cannot access a workstation USB reader by itself
Email	Supports Brevo HTTPS and Gmail/SMTP. The user reports Brevo; live provider selection, sender configuration, and delivery require verification
Browser-local storage	IndexedDB for supported queued actions and selected cached data; service worker for eligible application assets

Insights: Actual Implementation

Indicator	Current method
Seven-day attendance forecast	JavaScript Holt-Winters additive calculations with a bounded weekday baseline. Data volume/freshness checks identify ready or limited output
30-day attendance risk	Counts recorded unapproved absences in the preceding 30-day window, excluding approved leave. It is a historical indicator, not a prediction of the next 30 days
Unusual arrival detection	Modified Z-score using a median absolute deviation baseline and readiness checks
Biometric operational health	Summaries of recorded match/verification activity and evaluation records; not proof of scanner accuracy

3. Workflows and Operating Rules

Workflow	Implemented behavior
Login and recovery	CAPTCHA, credentials, account-status checks, email OTP, sessions, first-login password change, and employee password recovery. Designated tester accounts use OTP bypass
Attendance	Fingerprint time-in/time-out or authorized manual entries; validates times, chronology, and overlapping sessions
Corrections	Administrator password and reason, source indicators, before/after history, and employee notification; settled payroll restricts edits
Leave	Employee requests; administrator approval/rejection with schedule, attendance-conflict, and payroll checks
Cash advances	Request approval, cash-given confirmation, outstanding-balance tracking, and repayment through settled payroll
Payroll	Attendance hours and rates, additions, deductions, carried unpaid earnings, payslip summaries, and settlement records
Offline synchronization	Supported actions saved locally; session validation, administrator confirmation for attendance, server conflict checks, and duplicate prevention

4. Data, Security, and Recovery

Employee identifiers link accounts, attendance, leave, cash advances, payroll, and notifications. Settings, audit events, and biometric records support administration.

Area	Controls or procedure
Authentication	Salted scrypt hashes, hashed OTP/session tokens, five-hour sessions, HttpOnly cookies, production Secure flag, and logout revocation
API access	Server-side roles and ownership checks, CSRF protection, and administrator password confirmation for designated actions
Validation and throttling	Bounded inputs, business-rule checks, one-use OTP/reset exchanges, request-size limits, and IP-based request limits
Biometric storage	AES-256-GCM encrypted templates; raw samples processed into templates
Audit and notifications	Administrative/security events, traceable corrections, personal notices, and audit-retention indexing
Backup and recovery	JSON backup of stored collections; full restore into an empty database, or selective recovery of missing business records while preserving existing records

Full restoration returns records to the backup state: later attendance, account changes, and payment confirmations are absent unless recovered separately. Automatic absence processing may classify missing recent workdays as Absent. Reconcile attendance and actual payments before operational use; a restore does not reverse money already paid. Exported CSV/Excel reports are not database recovery backups.